



KTU NOTES

The learning companion.

**KTU STUDY MATERIALS | SYLLABUS | LIVE
NOTIFICATIONS | SOLVED QUESTION PAPERS**

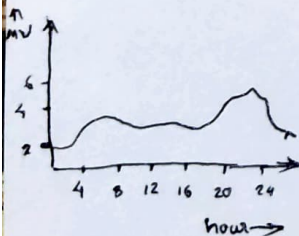
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Equations

1. Avg load = $\frac{\text{No. of unit generated in a day}}{24 \text{ hrs.}}$
2. Load factor = $\frac{\text{Avg load}}{\text{max demand}}$
3. Demand factor = $\frac{\text{max demand}}{\text{connected load}}$
4. Diversity factor = $\frac{\text{Sum of individual max load}}{\text{max demand on power station}}$
5. Plant capac. fac. = $\frac{\text{annual kWh o/p}}{\text{plant capacity} \times 8760 \text{ hrs}} = \frac{\text{Avg load}}{\text{P.C}}$
6. Reserved capacity = plant capacity - max demand.
7. plant use factor = $\frac{\text{Actual energy produced}}{\text{plant capac.} \times \text{hrs of use.}}$
8. Units genera./annum. = Avg load (kw) $\times$ hrs in a year
 $= \text{max demand} \times \text{L.F} \times 8760$

Load curve.

- shows variation of load on P.S w.r.t time.
- Load - P.S $\rightarrow$ not const. - varies w.r.t time
- load variation on whole day is recorded hourly and plotted against time on one graph. $\rightarrow$ known as daily load curve.

Importance.

1. shows variation of load.
2. area under curve gives $\rightarrow$ unit generated
3. Peak value in graph $\rightarrow$ max demand on that day
4. can be calc. Avg load = $\frac{\text{area under the curve (kWh)}}{24}$
5. helps in selecting plant size & no. of unit generated.
6. help to prepare operation sth.

Load duration curve.

- The variation on a P.S is graphically represented in descending magnitudes.
- It uses same data of L.C.
- area under L.C & L.D.C is equal.

Bulk power grid.

- Interconnected network for delivering electricity from producers to consumers.
- consist of 1. G.S, 2. Tr line, 3. Dis. line.

Micro power grid.

- modern, small scale version of the centralized electrical s/m
- Designed to provide power for a small community
- enables local power gen $\rightarrow$ local loads
- Highly flexible & efficient
- Can be connected $\rightarrow$ utility grid.
- Excess power can be sold $\rightarrow$ ↓

Need

- Answer to our energy crisis
- Tr losses ↓
- cuts emissions
- High quality & reliable

Components

- Radial feeder
- CHP & nonCHP micro sources
- controllers - central, micro source.
- CB
- sectionalizing CB
- storage device

Smart grid.

- Advanced digital 2-way power flow slm
- Capable of self-healing, adaptive, resilient & sustainable & have a prediction under diff. uncertainties
- Advanced cyber-security.

Components

- sensors & smart meters
- wireless comm.
- AI

Sag

$$W_i = 1 \text{ cc} \times \text{vol. of } 1 \text{ m}$$

$$= 1. \text{cc} \times \pi t (d+t) \times 100.$$

$$W_w = \text{Wind pressure} \times (\pi + 2t) \times 100.$$

$$T = \frac{\text{Ultimate stress} \times \text{conductor area}}{\text{S.F.}}$$

$$= \frac{\text{Working/ Breaking stress} \times \text{conductor area}}{\text{S.F.}}$$

$$C = \frac{2\pi\epsilon_0 \cdot \epsilon_r \cdot l}{\log_e\left(\frac{D}{d}\right)} \quad \text{F/m.}$$

$$I_c = \frac{V_{ph}}{X_c} = V_{ph} \cdot 2\pi f (C_e + 3C_c)$$

$$C_f = 3C_c.$$

$$C_e = \frac{C_f}{3}$$

$$g_{\max} = \frac{2V}{d \cdot \log_e\left(\frac{D}{d}\right)} \quad \text{volt/m} \quad \text{max stress} \rightarrow \text{cond surface.}$$

$$g_{\min} = \frac{2V}{D \cdot \log_e\left(\frac{D}{d}\right)}$$

MODULE-2.

1. Skin Effect.

- Tendency of alternating current to concentrate near the surface of the cond.

- DC $\rightarrow$ uniformly distn. over the whole cross section

- Effects

1. cross-sectional area of flow $\downarrow$
thus $\uparrow \rightarrow$ resistance.

$$R \propto \frac{\rho \cdot l \cdot f \cdot \mu}{\delta}$$

- Depends on.

1. Nature of material
2. Diameter of wire
3. frequency
4. shape of cond.

2. proximity effect.

Alternating m-flux in a cond. caused by the current flowing in a neighboring cond. gives rise to circulating current which has an apparent $\uparrow$ in resistance of a cond. $\rightarrow$ proximity effect

- Depends on

- cond. size
- frequency
- distance b/w cond.
- permeability of cond. material.

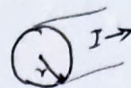
3. Ferranti Effect.

$$L = \frac{\Psi}{I} \text{ — flux linkage in wbl}$$

$$I \text{ — current.}$$

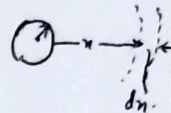
1. Internal flux.

$$\Psi_{int} = \frac{\mu_0 I}{8\pi} \text{ wbl turns / meter}$$



2. External flux.

$$\Psi_{ext} = \frac{\mu_0 I}{2\pi} \left[\frac{1}{4} + \int_r^\infty \frac{1}{x} \cdot dx \right] \text{ wbl turns/m.}$$



3. Single phase - 2 wire.

- Consider 1 ϕ OH line — 2 wbl cond.

- A & B of radius r & spaced d



- A & B carrying same current

$$\therefore I_A = -I_B \Rightarrow I_A + I_B = 0$$

- Flux linkages with cond. A. due to its own current I_A

$$= \frac{\mu_0 I_A}{2\pi} \left[\frac{1}{4} + \int_r^\infty \frac{1}{x} \cdot dx \right] \text{ — (1)}$$

- Flux linkages with cond. A due to current I_B .

$$= \frac{\mu_0 \cdot I_B}{2\pi} \int_d^\infty \frac{1}{x} \cdot dx. \text{ — (2)}$$

$\therefore$ Total flux linkages with cond. A. is,

$$\Psi_A = (1) + (2).$$

$$= \frac{\mu_0 I_A}{2\pi} \left[\frac{1}{4} + \int_r^\infty \frac{1}{x} \cdot dx \right] + \frac{\mu_0 I_B}{2\pi} \int_d^\infty \frac{1}{x} \cdot dx$$

$$= \frac{\mu_0}{2\pi} \left[\left(\frac{1}{4} + \int_r^\infty \frac{1}{x} \cdot dx \right) I_A + I_B \int_d^\infty \frac{1}{x} \cdot dx \right]$$

$$= \frac{\mu_0}{2\pi} \left[\left(\frac{1}{4} + \log_e \infty - \log_e r \right) I_A + I_B (\log_e \infty - \log_e d) \right]$$

$$= \frac{\mu_0}{2\pi} \left[\left(\frac{I_A}{4} + I_A \log_e \infty - I_A \log_e r + I_B \log_e \infty - I_B \log_e d \right) \right]$$

$$= \frac{\mu_0}{2\pi} \left[\frac{I_A + (I_A + I_B)}{4} \log_e \infty - I_A \log_e r - I_B \log_e d \right]$$

$$\Psi_A = \frac{\mu_0}{2\pi} \left[\frac{I_A}{4} - I_A \log_e r - I_B \log_e d \right]$$

$$\therefore I_A + I_B = 0 \Rightarrow -I_A = I_B.$$

$$\Psi_A = \frac{\mu_0}{2\pi} \left[\frac{I_A}{4} + I_A \log_e d - I_A \log_e r \right]$$

$$\Psi_A = \frac{\mu_0}{2\pi} \left[\frac{I_A}{4} + I_A \log_e \left(\frac{d}{r} \right) \right]$$

$$\Psi_A = \frac{\mu_0 I_A}{2\pi} \left[\frac{1}{4} + \log_e \left(\frac{d}{r} \right) \right] \text{ Wb turn/m}$$

$$L_A = \frac{\Psi_A}{I_A}, \text{ H/m}$$

$$L_A = \frac{\mu_0}{2\pi} \left[\frac{1}{4} + \log_e \left(\frac{d}{r} \right) \right] \text{ H/m}$$

$$= \frac{2 \times 10^{-7}}{2\pi} \left[\frac{1}{4} + \log_e \left(\frac{d}{r} \right) \right] \quad \mu_0 = 4\pi \times 10^{-7}$$

$$\rightarrow \left(\frac{1}{2} + 2 \log_e \left(\frac{d}{r} \right) \right) \times 10^{-7}$$

$$\text{Loop inductance} = 2 \cdot L_A.$$

$$= \left[1 + 4 \log_e \left(\frac{d}{r} \right) \right] \times 10^{-7}$$

Alternate form

$$L_A = 2 \times 10^{-7} \log_e \left(\frac{d}{r_1} \right) \text{ H/m} \rightarrow r_1 = 0.7788 \cdot r$$

$r_1 \Rightarrow$ GMR. $\rightarrow$ applicable to solid cond.

$$\text{loop inductance} = 2 \times L_A.$$

$$= 4 \times 10^{-7} \log_e \left(\frac{d}{r_1} \right) \text{ H/m}$$

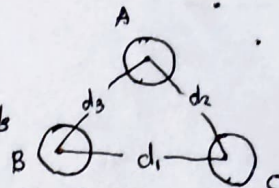
7. 3 ϕ - OH line.

- 3 cond. A, B, C.

- $I_A, I_B, I_C \rightarrow$ spacing d_1, d_2, d_3

- $A-B \Rightarrow d_3$ $B-C \Rightarrow d_1$ $C-A \Rightarrow d_2$

- Balanced, $I_A + I_B + I_C = 0$.



* Flux linkage with cond. A due to its own current I_A .

$$= \frac{\mu_0 I_A}{2\pi} \left[\frac{1}{4} + \int_r^\infty \frac{dn}{n} \right] \quad \text{--- (1)}$$

* Flux link. with cond. A due to I_B

$$= \frac{\mu_0 \cdot I_B}{2\pi} \int_{d_3}^\infty \frac{dn}{n} \quad \text{--- (2)}$$

* Flux link with cond. A due to I_C

$$= \frac{\mu_0 \cdot I_C}{2\pi} \int_{d_2}^\infty \frac{dn}{n} \quad \text{--- (3)}$$

$\therefore$ Total flux link with cond. A is,

$$\Psi_A = (1) + (2) + (3).$$

$$= \frac{\mu_0 I_A}{2\pi} \left[\frac{1}{4} + \int_r^\infty \frac{dn}{n} \right] + \frac{\mu_0 I_B}{2\pi} \int_{d_3}^\infty \frac{dn}{n} + \frac{\mu_0 I_C}{2\pi} \int_{d_2}^\infty \frac{dn}{n}$$

$$= \frac{\mu_0}{2\pi} \left[\left(\frac{I_A}{4} + I_A (\log_e \infty - \log_e r) + I_B (\log_e \infty - \log_e d_3) + I_C (\log_e \infty - \log_e d_2) \right) \right]$$

$$= \frac{\mu_0}{2\pi} \left[\frac{I_A}{4} + \log_e \infty [I_A + I_B + I_C] - I_A \log_e r - I_B \log_e d_3 - I_C \log_e d_2 \right]$$

$$\Psi_A = \frac{\mu_0}{2\pi} \left[\left(\frac{1}{4} - \log_e r \right) I_A - I_B \log_e d_3 - I_C \log_e d_2 \right]$$

a. Symmetrical spacing:

- A, B & C $\rightarrow$ sym. placed $\rightarrow$ like corners of Δ .

- $d_1 = d_2 = d_3$.

$$\therefore \Psi_A = \frac{\mu_0}{2\pi} \left[\left(\frac{1}{4} - \log_e r \right) I_A - I_B \log_e d - I_C \log_e d \right]$$

$$= \frac{\mu_0}{2\pi} \left[\left(\frac{1}{4} - \log_e r \right) I_A - (I_B + I_C) \log_e d \right]$$

$$= \frac{\mu_0}{2\pi} \left[\left(\frac{1}{4} - \log_e r \right) I_A + I_A \log_e d \right] \quad - I_A = I_B + I_C$$

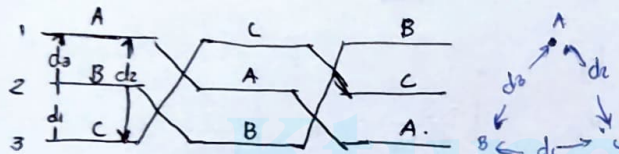
$$= \frac{\mu_0 I_A}{2\pi} \left[\frac{1}{4} + \log_e \left(\frac{d}{r} \right) \right] \quad \text{wb-turns/m.}$$

$$L_A = \frac{\Psi_A}{I_A}$$

$$= \frac{\mu_0}{2\pi} \left[\frac{1}{4} + \log_e \left(\frac{d}{r} \right) \right] \quad \mu_0 = 4\pi \times 10^{-7}$$

$$= \left[\frac{1}{2} + 2 \log_e \left(\frac{d}{r} \right) \right] \times 10^{-7} \text{ H/m}$$

b. Unsymmetrical spacing



- Let us assume each of the 3 sections is 1m length.

- Balanced cond. $I_A + I_B + I_C = 0$

- Let the line current be.

$$I_A = I(1 + j0)$$

$$I_B = I(-0.5 - j0.866)$$

$$I_C = I(-0.5 + j0.866)$$

From above

$$\Psi_A = \frac{\mu_0}{2\pi} \left[\left(\frac{1}{4} - \log_e r \right) I_A - I_B \log_e d_3 - I_C \log_e d_2 \right]$$

$$= \frac{\mu_0}{2\pi} \left[\left(\frac{1}{4} - \log_e r \right) I - I(-0.5 - j0.866) \log_e d_3 - I(-0.5 + j0.866) \log_e d_2 \right]$$

$$= \frac{\mu_0}{2\pi} \left[\frac{I}{4} - I \log_e r + 0.5 I \log_e d_3 + j0.866 I \log_e d_3 + 0.5 I \log_e d_2 - j0.866 I \log_e d_2 \right]$$

$$= \frac{\mu_0}{2\pi} \left[\frac{I}{4} - I \log_e r + 0.5 I \left[\log_e d_3 + \log_e d_2 \right] \right]$$

We know that,

$$\begin{aligned} 0.5 I (\log_e d_3 + \log_e d_2) &= 0.5 I \log_e (d_3 d_2) \\ &= I \log_e (d_3 d_2)^{0.5} \\ &= I \log_e \sqrt{d_3 d_2} \end{aligned}$$

$$j0.866 I (\log_e d_3 - \log_e d_2) = j0.866 I \log_e \left(\frac{d_3}{d_2} \right)$$

$$\therefore \Psi_A = \frac{\mu_0}{2\pi} \left[\frac{I}{4} - I \log_e r + I \log_e \sqrt{d_3 d_2} + j0.866 I \log_e \left(\frac{d_3}{d_2} \right) \right]$$

$$\Psi_A = \frac{\mu_0 I}{2\pi} \left[\frac{1}{4} + \log_e \frac{\sqrt{d_3 d_2}}{r} + j0.866 \log_e \left(\frac{d_3}{d_2} \right) \right]$$

$$\text{inductance } L_A = \frac{\Psi_A}{I_A} = \frac{\Psi_A}{I}$$

$$L_A = \frac{\mu_0}{2\pi} \left[\frac{1}{4} + \log_e \frac{\sqrt{d_3 d_2}}{r} + j0.866 \log_e \left(\frac{d_3}{d_2} \right) \right]$$

$$L_A = \left[\frac{1}{2} + 2 \log_e \frac{\sqrt{d_2 d_3}}{r} + 1.732 j \log_e \left(\frac{d_3}{d_2} \right) \right] \times 10^{-7}$$

Similarly,

$$L_B = \left[\frac{1}{2} + 2 \log_e \frac{\sqrt{d_3 d_1}}{r} + 1.732 j \log_e \left(\frac{d_1}{d_3} \right) \right] \times 10^{-7}$$

$$L_C = \left[\frac{1}{2} + 2 \log_e \frac{\sqrt{d_1 d_2}}{r} + 1.732 j \log_e \left(\frac{d_2}{d_1} \right) \right] \times 10^{-7}$$

$\therefore$ Induc. of each conductor =

$$= \frac{1}{3} (L_A + L_B + L_C)$$

$$= \left[\frac{1}{2} + 2 \log_e \sqrt[3]{\frac{d_1 d_2 d_3}{r}} \right] \times 10^{-7} \text{ H/m}$$

The distance 'd' is known as equivalent equilateral spacing for unsymmetrical transposed line

(i) Self GMD

$$\text{Inductance/cond./m} = 2 \times 10^{-7} \log_e \left(\frac{d}{D_s} \right)$$

$$D_s = \text{GMR or self GMD} = 0.7788 r$$

(ii) Mutual GMD.

Inductance formulas in terms of GMD.

(1) single phase line

$$\text{ind./cond./m} = 2 \times 10^{-7} \log_e \frac{D_m}{D_s}$$

$D_m \rightarrow$ spacing b/w cond $\rightarrow d$.

$D_s \rightarrow 0.7788 r$.

2. single ckt 3 ϕ line

$$\text{ind./phase/m} = 2 \times 10^{-7} \log_e \frac{D_m}{D_s}$$

$$D_m \rightarrow (d_1 \cdot d_2 \cdot d_3)^{1/3}$$

$D_s \rightarrow 0.7788 r$.

3. double ckt 3 ϕ line

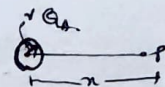
$$\text{ind./phase/m} = 2 \times 10^{-7} \log_e \frac{D_m}{D_s}$$

$$D_m \rightarrow (D_{AB} \cdot D_{BC} \cdot D_{CA})^{1/3}$$

$$D_s \rightarrow (D_{s1} \cdot D_{s2} \cdot D_{s3})^{1/3}$$

capacitance.

1. potential at a charged 1 cond.



$$V_A = \frac{Q_A}{2\pi\epsilon_0} \int \frac{1}{x} dx$$

2. potential at a charge cond. in a group of cond's

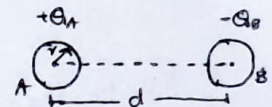
$$V_A = \int \frac{Q_A}{2\pi\epsilon_0 x} dx + \int \frac{Q_B}{2\pi\epsilon_0 x_1} dx + \int \frac{Q_C}{2\pi\epsilon_0 x_2} dx + \dots$$

$$Q_A + Q_B + Q_C = 0$$

$$= \frac{1}{2\pi\epsilon_0} \left[Q_A \log_e \left(\frac{1}{r} \right) + Q_B \log_e \left(\frac{1}{d_1} \right) + Q_C \log_e \left(\frac{1}{d_2} \right) + \dots \right]$$

1. single phase - 2 wire

- 2 parallel cond.



$\therefore$ Total pd b/w cond A & neutral 'infinite' plane is

$$V_A = \int \frac{Q}{2\pi x \epsilon_0} dx + \int \frac{-Q}{2\pi x \epsilon_0} dx$$

$$= \frac{Q}{2\pi \epsilon_0} \left[\int \frac{1}{x} dx - \int \frac{1}{x} dx \right]$$

$$= \frac{Q}{2\pi \epsilon_0} \left[(\log_e \infty - \log_e r) - (\log_e \infty - \log_e d) \right]$$

$$= \frac{Q}{2\pi \epsilon_0} \left[\log_e \left(\frac{\infty}{r} \right) - \log_e \left(\frac{\infty}{d} \right) \right] \text{ Volts.}$$

$$V_A = \frac{Q}{2\pi \epsilon_0} \cdot \log \left(\frac{d}{r} \right) \text{ volts}$$

$$\text{Similarly, } V_B = \int \frac{-Q}{2\pi x \epsilon_0} dx + \int \frac{Q}{2\pi x \epsilon_0} dx$$

$$V_B = \frac{-Q}{2\pi \epsilon_0} \log \left(\frac{d}{r} \right) \text{ volts}$$

$$V_{AB} = V_A - V_B \Rightarrow \frac{Q}{2\pi \epsilon_0} \log \left(\frac{d}{r} \right) - \left(-\frac{Q}{2\pi \epsilon_0} \log \left(\frac{d}{r} \right) \right)$$

$$= 2 \cdot \frac{Q}{2\pi \epsilon_0} \log \left(\frac{d}{r} \right)$$

$$= \frac{Q}{\pi \epsilon_0} \cdot \log \left(\frac{d}{r} \right)$$

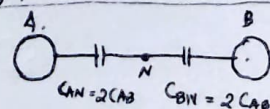
$$C_{AB} = \frac{Q}{V_{AB}} = \frac{Q}{\frac{Q}{\pi \epsilon_0} \cdot \log \left(\frac{d}{r} \right)}$$

$$= \frac{\pi \epsilon_0}{\log \left(\frac{d}{r} \right)} \text{ F/m.}$$

$\therefore$ Capa. to neutral,

$$C_N = C_{AN} = C_{BN} = 2C_{AB}$$

$$C_N = 2 \cdot \frac{\pi \epsilon_0}{\log \left(\frac{d}{r} \right)} \text{ F/m} \Rightarrow \text{only for solid ground conductor.}$$



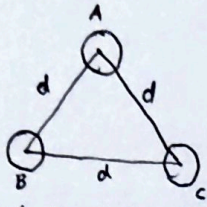
1. Capacitance of 3φ OH line

- In 3φ line - capacitance of each conductor is considered instead of capacitance from cond. to cond.

Two cases -

(1) Symmetrical spacing

- 3 cond. A, B, C.
- charges Q_A, Q_B, Q_C / m length
- Capac. from line cond. to neutral is need to find.
- overall p.d. b/w cond. A & infinite neutral plane is



$$V_A = \int \frac{Q_A}{2\pi r \epsilon_0} dr + \int \frac{Q_B}{2\pi r \epsilon_0} dr + \int \frac{Q_C}{2\pi r \epsilon_0} dr$$

$$= \frac{1}{2\pi \epsilon_0} \left[Q_A (\log_e \infty - \log_e \frac{1}{r}) + Q_B (\log_e \infty - \log_e d) + Q_C (\log_e \infty - \log_e d) \right]$$

$$-\log_e r \Rightarrow \log_e(r^{-1}) \Rightarrow \log_e\left(\frac{1}{r}\right)$$

$$\therefore = \frac{1}{2\pi \epsilon_0} \left[Q_A \cdot \log_e\left(\frac{1}{r}\right) + (Q_B + Q_C) \log_e\left(\frac{1}{d}\right) + (Q_A + Q_B + Q_C) \log_e \infty \right]$$

$$= \frac{1}{2\pi \epsilon_0} \left[Q_A \log_e \frac{1}{r} - Q_A \log_e\left(\frac{1}{d}\right) \right]$$

$$V_A = \frac{Q_A}{2\pi \epsilon_0} \cdot \log_e\left(\frac{d}{r}\right) \text{ volts}$$

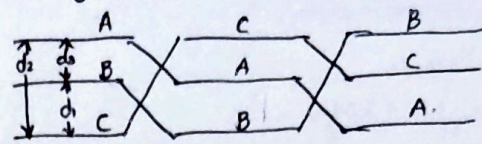
$$Q_B + Q_C = -Q_A$$

$\therefore$ Capacitance of A w.r.t neutral.

$$C_A = \frac{Q_A}{V_A} = \frac{Q_A}{\frac{Q_A}{2\pi \epsilon_0} \cdot \log_e\left(\frac{d}{r}\right)} = \frac{2\pi \epsilon_0}{\log_e\left(\frac{d}{r}\right)} \text{ F/m}$$

It is similar to C_B & C_C .

(2) Unsymmetrical spacing



Let us assume balanced conditions

$$Q_A + Q_B + Q_C = 0$$

Considering all the 3 sections of the transposed line for phase A,

1st position

$$V_1 = \left[Q_A \log_e\left(\frac{1}{r}\right) + Q_B \log_e\left(\frac{1}{d_3}\right) + Q_C \log_e\left(\frac{1}{d_2}\right) \right]$$

2nd position

$$V_2 = \frac{1}{2\pi \epsilon_0} \left[Q_A \log_e\left(\frac{1}{r}\right) + Q_B \log_e\left(\frac{1}{d_1}\right) + Q_C \log_e\left(\frac{1}{d_3}\right) \right]$$

3rd position

$$V_3 = \frac{1}{2\pi \epsilon_0} \left[Q_A \log_e\left(\frac{1}{r}\right) + Q_B \log_e\left(\frac{1}{d_2}\right) + Q_C \log_e\left(\frac{1}{d_1}\right) \right]$$

$\therefore$ Avg voltage on conductor A

$$V_A = \frac{1}{3} (V_1 + V_2 + V_3)$$

on solving,

$$V_A = \frac{1}{3} \times \frac{1}{2\pi \epsilon_0} \left[Q_A \cdot \log_e\left(\frac{1}{r}\right) + (Q_B + Q_C) \log_e\left(\frac{1}{d_1 d_2 d_3}\right) \right]$$

$$Q_A + Q_B + Q_C = 0; Q_B + Q_C = -Q_A$$

$$V_A = \frac{Q_A}{6\pi \epsilon_0} \left[\log_e\left(\frac{d_1 d_2 d_3}{r^3}\right) \right] \quad \frac{1}{r^3}$$

$$\text{OR} \quad V_A = \frac{Q_A}{2\pi \epsilon_0} \log_e\left(\frac{d_1 d_2 d_3}{r}\right)^{\frac{1}{3}}$$

$\therefore$ Capacitance from cond. to neutral.

$$C_A = \frac{Q_A}{V_A} = \frac{Q_A}{\frac{Q_A}{2\pi \epsilon_0} \log_e\left(\frac{d_1 d_2 d_3}{r}\right)^{\frac{1}{3}}} = \frac{2\pi \epsilon_0}{\log_e\left(\frac{d_1 d_2 d_3}{r}\right)^{\frac{1}{3}}}$$

Classification of OH line

1. short tr line

- 50 km ; < 20 kv
- capacitance effect is very small & neglected
- Resistance & inductance is considered.

2. Medium tr line

- 50-150 km ; 20-100 kv
- capacitance effect are considered.
-

3. long tr line

- > 150 km ; > 150 kv
- line constants are uniformly distributed over whole length.

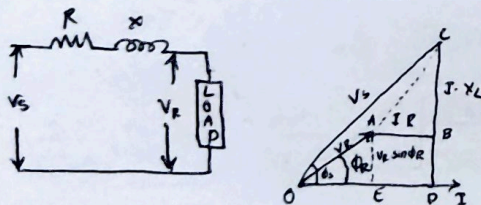
Voltage Regulation

$$V_R = \frac{V_S - V_R}{V_R} \times 100.$$

Transmission Efficiency

$$\eta = \frac{V_R \cdot I_R \cdot \cos \phi_R}{V_S \cdot I_S \cdot \cos \phi_S} \times 100.$$

Performance of 1φ short tr line.



$$V_S = \sqrt{(V_R \cos \phi_R + I_R R)^2 + (V_R \sin \phi_R + I_R X_L)^2}$$

$$\% V_R = \frac{V_S - V_R}{V_R} \times 100$$

$$\text{Sending end p.f. } \cos \phi_S = \frac{OP}{OC} = \frac{V_R \cos \phi_R + I_R R}{V_S}$$

$$\text{Power delivered} = V_R \cdot I_R \cos \phi_R$$

$$\text{Line losses} = I^2 R$$

$$\text{Power sent out} = V_R \cdot I_R \cos \phi_R + I^2 R$$

$$\begin{aligned} \text{Lagging } + \\ \text{Leading } - \end{aligned} \quad V_S = V_R + I_R \cos \phi_R + I \cdot X_L \sin \phi_R$$

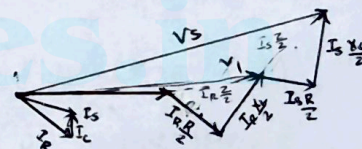
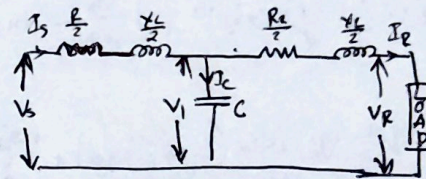
3φ short tr line

$$\% V_R = \frac{I_R \cos \phi_R + I X_L \sin \phi_R}{V_R} \times 100 \quad (\text{lagging})$$

$$= \frac{I_R \cos \phi_R - I X_L \sin \phi_R}{V_R} \quad (\text{leading})$$

Medium tr lines

1. Nominal T method.



$$\bar{V}_R = (V_R \angle 0)$$

$$\bar{I}_R = I_R (\cos \phi_R - j \sin \phi_R)$$

$$V_{C1} = \bar{V}_R + \bar{I}_R \frac{Z}{2}$$

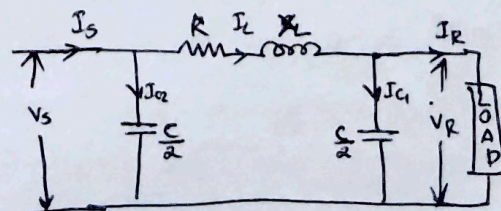
$$I_C = V_{C1} \cdot j\omega C$$

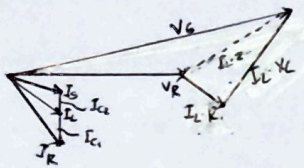
$$I_S = I_C + I_R$$

$$V_S = V_1 + I_S \frac{Z}{2}$$

$$Z = \left(\frac{R}{2} + j \frac{X_L}{2} \right)$$

2. Nominal Pi method.





$$V_R = V_R + j0$$

$$I_R = I_R (\cos \phi_R - j \sin \phi_R)$$

$$I_{C1} = j\omega C \frac{V_R}{2}$$

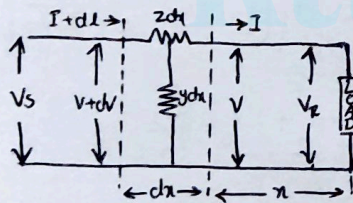
$$I_L = I_R + I_{C1}$$

$$I_{C2} = j\omega C \frac{V_S}{2}$$

$$V_S = V_R + I_L \cdot Z$$

$$I_S = I_L + I_{C2}$$

Long tr lines



Z-series impedance of line (unit length)

Y dx → shunt imp

I + dI → considering the element dx
V + dV → Vol. at the end of element toward sending end.

I → current leaving the element dx.

for small element dx,

Z dx → series impedance

Y dx → shunt admittance.

$$\frac{dV}{dx} = I \cdot Z \quad \text{--- (1)}$$

$$\frac{dI}{dx} = V \cdot Y \quad \text{--- (2)}$$

Diff (1) w.r.t x.

$$\frac{d^2 V}{dx^2} = Z \cdot \frac{dI}{dx}$$

$$\frac{d^2 V}{dx^2} = Z \cdot V \cdot Y \quad \text{--- (3)}$$

$$A = \cosh \sqrt{YZ}$$

$$B = \sqrt{\frac{Z}{Y}} \sinh \sqrt{YZ}$$

$$C = \sqrt{\frac{Y}{Z}} \sinh \sqrt{YZ}$$

$$D = \cosh \sqrt{YZ}$$

The soln. of this diff equ. is

$$V = k_1 \cosh (\alpha \sqrt{YZ}) + k_2 \sinh (\alpha \sqrt{YZ}) \quad \text{--- (4)}$$

Diff (4) w.r.t x.

$$\left(\frac{dV}{dx} \right)^{I_Z} = k_1 \sqrt{YZ} \sinh (\alpha \sqrt{YZ}) + k_2 \sqrt{YZ} \cosh (\alpha \sqrt{YZ})$$

$$\therefore I \cdot Z = k_1 \sqrt{YZ} \sinh (\alpha \sqrt{YZ}) + k_2 \sqrt{YZ} \cosh (\alpha \sqrt{YZ})$$

$$I = \sqrt{\frac{Y}{Z}} \left[k_1 \sinh (\alpha \sqrt{YZ}) + k_2 \cosh (\alpha \sqrt{YZ}) \right] \quad \text{--- (5)}$$

At $x=0$, $V=V_R$ & $I=I_R$. --- (a)

putting (a) in (4).

$$V_R = k_1 \cosh 0 + k_2 \sinh 0 = k_1 + 0$$

$$\underline{V_R = k_1}$$

putting (a) in (5).

$$I = \sqrt{\frac{Y}{Z}} \left[k_1 \sinh 0 + k_2 \cosh 0 \right] = \sqrt{\frac{Y}{Z}} (0 + k_2)$$

$$\therefore \underline{k_2 = \sqrt{\frac{Z}{Y}} \cdot I_R}$$

Subs values of k_1 & k_2 in (4) & (5).

$$V = V_R \cosh (\alpha \sqrt{YZ}) + \sqrt{\frac{Z}{Y}} I_R \sinh (\alpha \sqrt{YZ}) \quad \text{--- (6)}$$

$$I = \sqrt{\frac{Y}{Z}} \left[V_R \sinh \alpha \sqrt{YZ} + I_R \cosh (\alpha \sqrt{YZ}) \right] \quad \text{--- (7)}$$

put $x=l$; $V=V_S$; $I=I_S$ in (6) & (7).

$$V_S = V_R \cosh (\alpha \sqrt{YZ}) + \sqrt{\frac{Z}{Y}} I_R \sinh \alpha \sqrt{YZ}$$

$$I_S = \sqrt{\frac{Y}{Z}} \left[V_R \sinh \alpha \sqrt{YZ} + \sqrt{\frac{Z}{Y}} I_R \cosh \alpha \sqrt{YZ} \right]$$

$$I \sqrt{YZ} = \sqrt{I_Y \cdot I_Z} = \sqrt{YZ} \quad Y \rightarrow \text{total shunt admittance}$$

$$\sqrt{\frac{Y}{Z}} = \sqrt{\frac{Y I}{Z I}} = \sqrt{\frac{Y}{Z}} \quad Z \rightarrow \text{total series impedance}$$

$$V = V_R \cosh \sqrt{YZ} + \sqrt{\frac{Z}{Y}} I_R \sinh \sqrt{YZ}$$

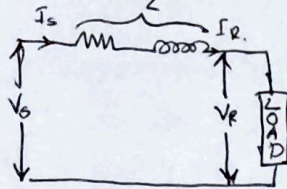
$$I = V_R \sqrt{\frac{Y}{Z}} \sinh \sqrt{YZ} + I_R \cosh \sqrt{YZ}$$

Generalised ckt constant of a Tr line

$$\bar{V}_S = \bar{A} \cdot \bar{V}_R + \bar{B} \cdot \bar{I}_R$$

$$\bar{I}_S = \bar{C} \bar{V}_R + \bar{D} \bar{I}_R$$

1. Short line



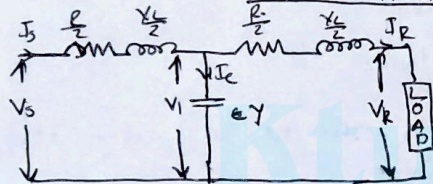
$$\bar{V}_S = \bar{V}_R + \bar{I}_R \bar{Z} \quad \therefore \bar{A} = 1 \quad \bar{B} = \bar{Z}$$

$$\bar{I}_S = \bar{I}_R \quad \bar{C} = 0 \quad \bar{D} = 1$$

$$\bar{A} = \bar{D}; \quad \bar{A}\bar{D} - \bar{B}\bar{C} = 1$$

2. Medium lines

Nominal T method.



$$\bar{V}_S = \bar{V}_1 + \bar{I}_S \frac{\bar{Z}}{2} \quad \bar{V}_1 = \bar{V}_R + \bar{I}_R \frac{\bar{Z}}{2}$$

$$\bar{I}_R = \bar{I}_S - \bar{I}_C = \bar{V}_1 \cdot \bar{Y}$$

$$\bar{I}_C = \bar{V}_1 \cdot \bar{Y}$$

$$= \bar{Y} \left[\bar{V}_R + \bar{I}_R \frac{\bar{Z}}{2} \right]$$

$$\bar{I}_S = \bar{I}_R + \bar{I}_C$$

$$= \bar{I}_R + \bar{Y} \left[\bar{V}_R + \bar{I}_R \frac{\bar{Z}}{2} \right]$$

$$\bar{Y} \bar{V}_R + \left(1 + \frac{\bar{Y}\bar{Z}}{2} \right) \bar{I}_R$$

$$\bar{V}_S = \bar{V}_1 + \bar{I}_S \frac{\bar{Z}}{2}$$

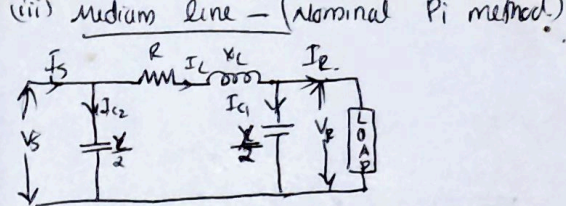
$$= \bar{V}_R + \bar{I}_R \frac{\bar{Z}}{2} + \bar{I}_S \frac{\bar{Z}}{2}$$

$$= \bar{V}_R + \bar{I}_R \frac{\bar{Z}}{2} + \bar{I}_R + \bar{Y} \left[\bar{V}_R + \bar{I}_R \frac{\bar{Z}}{2} \right] \frac{\bar{Z}}{2}$$

$$= \bar{V}_R + \bar{I}_R \frac{\bar{Z}}{2} + \bar{I}_R + \bar{Y} \frac{\bar{Z}}{2} \bar{V}_R + \frac{\bar{Y}\bar{I}_R \bar{Z}^2}{4}$$

$$= \left[1 + \frac{\bar{Y}\bar{Z}}{2} \right] \bar{V}_R + \left[2 + \frac{\bar{Y}\bar{Z}^2}{4} \right] \bar{I}_R$$

$$\begin{bmatrix} \bar{V}_S \\ \bar{I}_S \end{bmatrix} = \begin{bmatrix} \left(1 + \frac{\bar{Y}\bar{Z}}{2} \right) & \bar{Z} \left(1 + \frac{\bar{Y}\bar{Z}}{4} \right) \\ \bar{Y} & \left(1 + \frac{\bar{Y}\bar{Z}}{2} \right) \end{bmatrix} \begin{bmatrix} \bar{V}_R \\ \bar{I}_R \end{bmatrix}$$



$$\bar{V}_R = \bar{Z} = R + jX_L \quad \text{--- series impedance/phase}$$

$$\bar{Y} = j\omega C \quad \text{--- shunt impedance}$$

$$\bar{I}_S = \bar{I}_L + \bar{V}_S \cdot \frac{\bar{Y}}{2} \quad \bar{I}_L = \bar{I}_R + \bar{I}_C$$

$$= \bar{I}_R + \bar{V}_R \cdot \frac{\bar{Y}}{2}$$

$$\bar{V}_S = \bar{V}_R + \bar{I}_L \bar{Z}$$

$$= \bar{V}_R + \left(\bar{I}_R + \bar{V}_R \cdot \frac{\bar{Y}}{2} \right) \bar{Z}$$

$$= \left[1 + \frac{\bar{Y}\bar{Z}}{2} \right] \bar{V}_R + \bar{Z} \cdot \bar{I}_R$$

$$\bar{I}_S = \bar{I}_L + \bar{V}_S \cdot \frac{\bar{Y}}{2}$$

$$= \bar{I}_R + \left[\left(1 + \frac{\bar{Y}\bar{Z}}{2} \right) \bar{V}_R + \bar{Z} \cdot \bar{I}_R \right] \frac{\bar{Y}}{2}$$

$$= \bar{I}_R + \bar{V}_R \frac{\bar{Y}}{2} + \left[\left(1 + \frac{\bar{Y}\bar{Z}}{2} \right) \bar{V}_R + \bar{Z} \cdot \bar{I}_R \right] \frac{\bar{Y}}{2}$$

$$= \left[\bar{Y} + \frac{\bar{Z}\bar{Y}^2}{2} \right] \bar{V}_R + \left(1 + \frac{\bar{Y}\bar{Z}}{2} \right) \bar{I}_R$$

$$\begin{bmatrix} \bar{V}_S \\ \bar{I}_S \end{bmatrix} = \begin{bmatrix} \left(1 + \frac{\bar{Y}\bar{Z}}{2} \right) & \bar{Z} \\ \bar{Y} \left[\bar{Y} + \frac{\bar{Z}\bar{Y}^2}{2} \right] & \left(1 + \frac{\bar{Y}\bar{Z}}{2} \right) \end{bmatrix} \begin{bmatrix} \bar{V}_R \\ \bar{I}_R \end{bmatrix}$$

Sag:

- The diff. in level b/w points of supports & lower point on the cond.

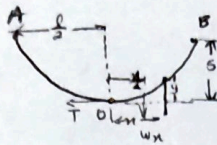


1. Supports are at equal levels

$l \rightarrow$ length of span

$w \rightarrow$ weight of cond

$T \rightarrow$ Tension in the cond.



- Consider a point P $\Rightarrow (x, y)$

- Two forces acting $\rightarrow$ 1. The weight w_x of cond acting at a distance $\frac{x}{2}$ from O.

Eqn. two forces about 2. The tension T acting at O.

$$T_y = w_x \cdot \frac{x}{2}$$

$$y = \frac{w x^2}{2T}$$

At support A; $x = \frac{l}{2}$; $y = S$

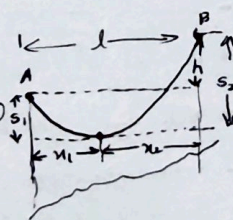
$$\text{Sag, } S = \frac{w (\frac{l}{2})^2}{2T} = \frac{w l^2}{8T}$$

2. Support are at unequal levels

$$S_1 = \frac{w x_1^2}{2T}$$

$$S_2 = \frac{w x_2^2}{2T}$$

$$\begin{cases} x_1 + x_2 = l \\ S_2 - S_1 = h \end{cases}$$



$$S_2 - S_1 = \frac{w x_2^2}{2T} - \frac{w x_1^2}{2T} \Rightarrow \frac{w}{2T} (x_2^2 - x_1^2)$$

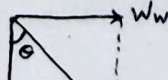
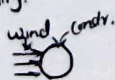
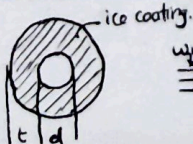
$$= \frac{w}{2T} (x_2 + x_1)(x_2 - x_1) \Rightarrow \frac{w l}{2T} (x_2 - x_1)$$

$$h = \frac{w l}{2T} (x_2 - x_1)$$

$$(x_2 - x_1) = \frac{2Th}{wl} \quad \text{Solving (1) \& (2)}$$

$$x_1 = \frac{l}{2} - \frac{Th}{wl} \quad x_2 = \frac{l}{2} + \frac{Th}{wl}$$

Effect of wind & ice loading



$$(w + w_i) \dots \dots \dots w_e$$

Total weight of cond / unit length

$$w_e = \sqrt{(w + w_i)^2 + (w_w)^2}$$

$w \rightarrow$ weight of cond

$w_i \rightarrow$ weight of ice / unit length

$w_w \rightarrow$ wind force / unit length

$$\tan \alpha = \frac{w_w}{w + w_i}$$

$$S = \frac{w_e \cdot l^2}{2T}$$

$$\text{Vertical Sag} = S \cdot \cos \alpha$$

$$\text{Working Tension, } T = \frac{\text{Ultimate strength}}{\text{Safety factor}}$$

$$\text{wt of cond / m length} = \text{Sp gravity} \times \text{vol. of 1m cond}$$

Insulator properties

- High mech strength - withstand loads
 - High elect resistance - avoid leakage currents to earth
 - High relative permability - Dielectric strength - high.
 - Should be non porous - otherwise $\downarrow$
 - High ratio of puncture strength to flashover.
- material - porcelain.

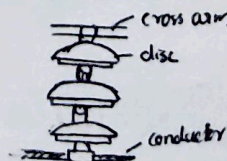
Types of insulators

1. Pin type: $\rightarrow$ upto 33 kV.



Safety factor - 10.

2. Suspension type. upto > 33 kV.



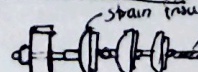
- one disc $\rightarrow$ 11 kV.

- 66 kV $\Rightarrow$ 6 disc.

- damaged disc $\rightarrow$ can replace.

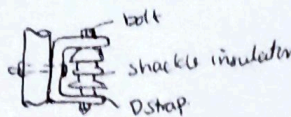
- provides partial prote. - lightning

3. strain insulators



< 11 kV

4. Shackle insulator



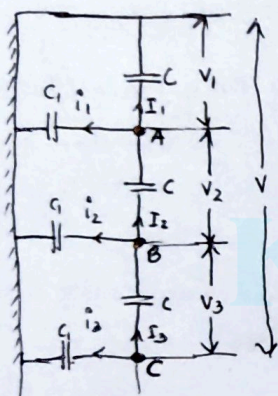
String Efficiency

- The ratio of vol across the whole string to the product of no. of discs and the vol. across the disc nearest to the conductor.

$$SE = \frac{\text{Vol. across the string}}{n \times \text{Vol. across disc nearest to and.}}$$

$n \rightarrow$ no. of disc

↑ SE - more uniform - vol. distribution



- Equivalent ckt - 3-disc string
- $C \rightarrow$ self capa.
- $C_1 = K \cdot C$
- Vol. across each unit V_1, V_2, V_3

Applying KCL to node A

$$I_2 = I_1 + i_1$$

$$V_2(\omega C) = V_1(\omega C) + V_1(\omega C) \cdot K$$

$$V_2 = V_1(1+K)$$

Applying KCL to node B

$$I_3 = I_2 + i_2$$

$$V_3\omega C = V_2\omega C + (V_1 + V_2)\omega K$$

$$V_3 = V_2 + (V_1 + V_2)K$$

$$= V_1K + (1+K)V_2 \quad V_2 = V_1(1+K)$$

$$= V_1K + (1+K)V_1(1+K)$$

$$= V_1K + V_1(1+K)^2 \Rightarrow K + 1 + 2K + K^2 = 1 + 3K + K^2$$

$$V_3 = V_1[1 + 3K + K^2]$$

Vol. b/w cond. A & B

$$V = V_1 + V_2 + V_3$$

$$V = V_1 + V_1(1+K) + V_1(1+3K+K^2)$$

$$V = V_1[3 + 4K + K^2]$$

$$V = V_1(1+K)(3+K)$$

$$\frac{V}{1} = \frac{V_2}{1+K} = \frac{V_3}{1+3K+K^2} = \frac{V}{(1+K)(3+K)}$$

$$\% SE = \frac{V}{3 \times V_3} \times 100$$

Methods to improve S.E

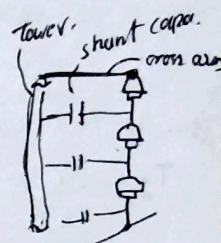
1. By using longer cross arm

- SE depends on K.

$$\Rightarrow \downarrow K = \uparrow SE$$

$$\Rightarrow \downarrow K = \downarrow \text{shunt capa}$$

$$\Rightarrow \downarrow \text{shunt} = \uparrow \text{cross arm length}$$



2. By grading the insulator

- Insulators - diff capacitance

- top - more ; bottom - more

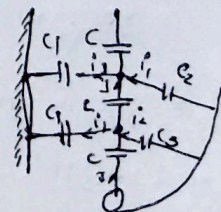
- Vol. $\frac{1}{\omega C}$ capacitance

- method $\rightarrow$ equilib the potential distri across the units in the string

3. By using guard ring

- potential across each unit can be equilized by using a guard ring.

- metal ring connec. to surrounding the bottom insulator.



Corona

- The phenomenon of violet glow, hissing sound and pdr of ozone gas in an overhead line.

Factors affecting corona.

1. Atmosphere - on stormy weather - corona $\uparrow$
2. Cond. size - shape & condition - Rough & irregular $\uparrow$
3. Spacing b/w cond - spacing $\uparrow$ - corona $\downarrow$
4. Line vol. - $\uparrow$ with corona.

Advantages.

- air surrounding of the cond - conducting
- virtual dia $\uparrow$
- dia $\uparrow$ - $\downarrow$ electro-static stress
- Vres - effect of transients produced by surges

Disadvantages.

- loss of energy $\rightarrow$ affects its efficiency
- Ozone production - causes corrosion
- causes interference with neighbouring commu line

Methods to reduce.

1. $\uparrow$ ing cond. size.
2. $\uparrow$ ing cond. spacing

• Critical disruptive vol.

- min phase to neutral vol. at which corona occurs

$$V_c = m_0 \cdot g_0 \cdot \delta \cdot r \cdot \log_e \frac{d}{r} \text{ kV/phase}$$

$g_0 =$

$m_0 = 1 \rightarrow$ polished

0.98-0.92 - dirty

0.87-0.8 - stranded.

• Visual - critical vol.

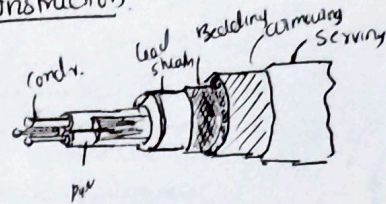
- min phase-neutral vol. at which corona glow appears along the line cond.

$$V_v = m_v \cdot g_0 \cdot \delta \cdot r \cdot \left(1 + \frac{0.3}{r \delta V}\right) \log_e \frac{d}{r}$$

Underground cables

- one or more cond. covered with suitable insulation and surrounded by a protective cover.

Construction.



1. Core

- consist 2 or more
- tinned Cu or Al
- stranded.

2. Insulation

- Each core provided - insulation
- Impregnated paper, varnished cambric, rubber mineral
- Thickness - voltage rating

3. Metallic sheath.

- To protect the cable from moisture, gases or other damaging liquids.
- Al or lead

4. Bedding

- over the metallic sheath
- consist fibrous material
- protection against corrosion

5. Armouring

- over bedding
- 1 or 2 layers of Galvanised steel or steel tape.
- protect from mechanical injury

6. Serving

- protect armouring from atm condition
- fibrous material - similar to bedding

Types of cables

LT - 1000V

HT - 11 kV

ST - 22-33 kV

EHT - 33-66 kV

EST. : > 132 kV

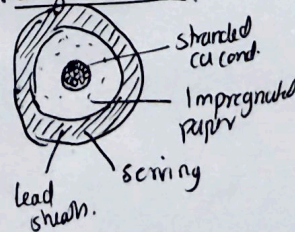
1. Type of insul. material

2. voltage rating

3. core type.

- single core
- two core
- 3 core
- four core

1. Single core LT.



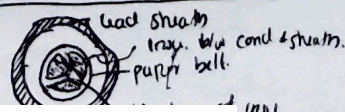
- upto 6600V

- tinned Al or Cu

- lead sheath.

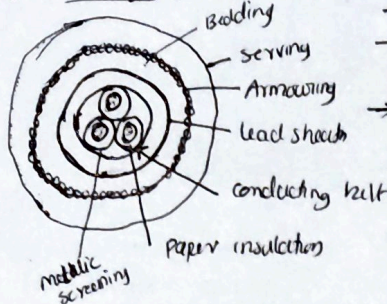
2. 3 core cables

1. Belted cable - upto 11 kV



2. Screened cables upto 33 KV

(i). H-type:



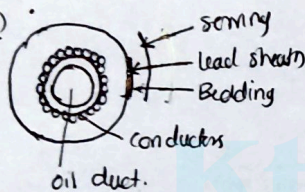
- Core insu. - impreg paper
- each core - covered - metallic screening - Al foil
- Additional conducting belt

(ii). S-L Type:

- Each core covered by its own lead sheath.
- There is no overall armoring & serving
- minimise core-core breakdown
- bending - easy.

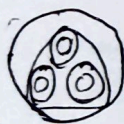
3. Pressure cables > 66 KV.

(oil)



- oil under pressure constantly supplied to the channel by external reservoirs.

(gas)



- ↑ pressure - tends to close voids
- All section & weight, low thermal expansion
- maintenance cost small

- No. 2 → help to quenching flame is any.
- cost ↑.

Insulation resistance.

$$R = \frac{\rho}{2\pi l} \log_e \frac{r_2}{r_1}$$

$r_2 \rightarrow$ radius - cable

$r_1 \rightarrow$ radius conductor.

Capacitance.

$$C = \frac{\epsilon_r \cdot l}{41.4 \log_{10} \left(\frac{D}{d} \right)} \times 10^{-9} \text{ F}$$

- process of achieving uniform electrostatic stress in the dielectric cables.

1. Capacitance grading

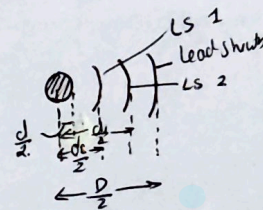
The process of achieving uniformity in the dielectric stress by using different layers of different dielectrics

$$V_1 = \frac{q_{max}}{2} d \log \frac{d_1}{a} \quad V_2 = \frac{q_{max}}{2} d_1 \log \frac{d_2}{d_1}$$

$$V_3 = \frac{q_{max}}{2} d_2 \log \frac{D_0}{d_2}$$

2. Intersheath grading

- homogeneous dielectric is used
- it is divided into various layers
- placed b/w core & lead sheath.
- vol. distn in dielectric is improved.
- uniform potential gradient is obtained



$$g_{1max} = \frac{V_1}{\frac{d_1}{2} \log \frac{d_1}{a}}$$

$$g_{2max} = \frac{V_2}{\frac{d_2}{2} \log \frac{d_2}{d_1}}$$

$$g_{3max} = \frac{V_3}{\frac{d_3}{2} \log \frac{D_0}{d_3}}$$

Surge Impedance Loading

- Essential parameter in P.S
- Used to determine max. loading capacity of tr line
- In long tr line > 250 km, when line is charged.
- capa → reactive power → Induc. component.
- If we take balance of two

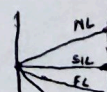
$$\text{Capacitive VAR} = \text{Inductive VAR}$$

$$\frac{V^2}{X_C} = I^2 X_L$$

$$\frac{V}{I} = \sqrt{X_C \cdot X_L} = \sqrt{\frac{2\pi f L l}{2\pi f C l}} = \sqrt{\frac{L}{C}}$$

$$\text{i.e. } Z_0 = \sqrt{\frac{L}{C}}$$

- SIL is defined as power delivered by a line to a purely resistive load equal in value to si of that line.



- Independent of length of line

⑥. FACTS - flexible AC Tr. S/m

- AC Tr. s/m incorporating power electronic based and other static controllers to enhance controllability & res power transfer capability.

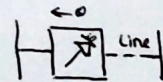
Benefits.

1. Control of power flow - by control - imped, angle / volt
2. res loading capability
3. res s/m security
4. prevent blackouts
5. Reduce reactive power flow
6. Upgrade of lines
7. Dampout power s/m oscillations

Classification

1. Series Controllers

- variable imp such as capacitor, reactor etc. or power electronic based variable source connected in series with the line to serve the requirement
- Series controllers inject voltage in series with the line
- variable imp $\times$ current flow through it, = injected series voltage.
- It supplies or consumes variable reactive power



Advantages

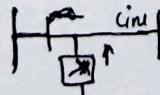
- steady state voltage regulation
- Dynamic voltage stabilization
- Improved transient stability
- Improved power trans

Example

1. TSSC
2. TC SC
3. static sync series compensator (SSSC)

2. Shunt controllers

- variable imp, variable source or combination of these connected across the line.
- Inject current into the s/m
- shunt imp connected to line voltage causes a current flow $\rightarrow$ current injection.



- Line voltage quadratures with current.

It consumes or supplies reactive power.

- other phase relationship - handling real power

Advantages.

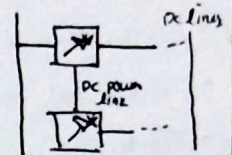
- steady state voltage
- Dynamic vol. stabilization
- stability improvement
- Dynamic load balancing

Example

1. TC R^{reactive}
2. TSC^{switched capacitor}
3. TCT^{thyristor controlled reactor}

Combined Series-series controllers.

- combination of separate series controllers which are controlled
- provide independent series reactive compensation for each line
- Also transfer real power among the line via power link
- manage real & reactive power flow



Advantages.

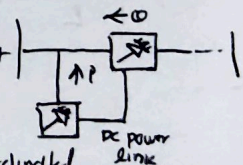
1. power balance in multi-line Tr s/m
2. Improved power transfer
3. simultaneous control of ||l lines

Example

- Interline power flow controller

Series-shunt controllers.

- combination of separate shunt & series controllers
- which are controlled in a coordinated manner.
- shunt part - inject current
- series part - inject voltage
- They can be unified and can transfer real power via power link



Advantages

- general adv. of shunt & series
- control of p.f in complex ps

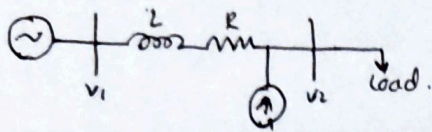
Example

- unified power flow controller

Compensation

- modification of electrical chara of the tr line in order to ↑ power transfer capability while satisfying the fundamental requirements for transmission

1. Shunt Compensation

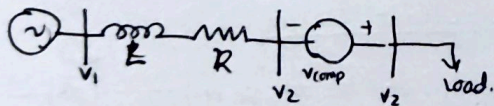


- inject current
- static syn compensator ; VAR compensator.
- provide vol. support
- Effective in maintaining desired vol.
- ineffective in controlling actual real power

Objectives

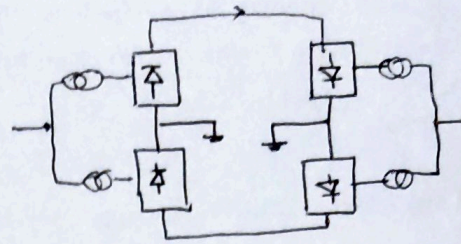
- ↑ transmittable power
- ↑ transient stability
- ↑ sm stability
- Reduce loss

2. Series Compensation

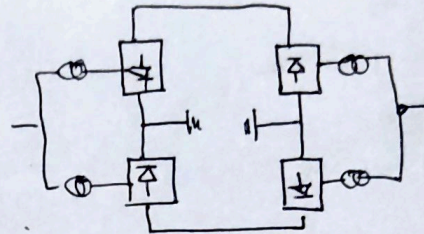


- controlling power flow

2. Bipolar

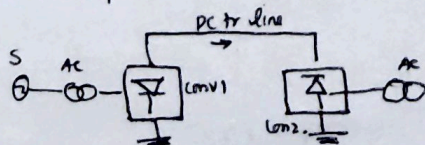


3. Homopolar



Types of DC links

1. monopolar link



MODULE - 4.

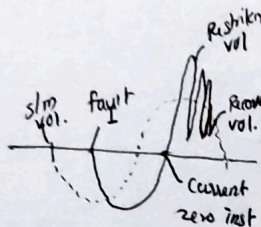
CB

automatically operated electrical switch designed to protect an elec ckt from sc / over load.

Formation of Arc

- Sc occurs, contact begins to separate
- Pel blow contact sufficient - ionise air - act as conductor - Arc struck b/w them.
- Depends on
 1. Degree of ionization
 2. Arc resistance
 3. Length of arc.
 4. Cross section of arc

$$R = \int \frac{\rho}{A}$$



Terms

ARC voltage

- Vol- appears across the contact of CB during arcing period.

Restriking voltage

- Transient vol. appears b/w contact at or near current zero during arcing period

Recovery

- Normal frequency rms voltage after arc extinction

Arc interruption method.

High resistance

Arc diss. red by

1. cooling of arc
 2. ↑ length of arc
 3. ↓ cross section of arc
 4. Splitting of arc.
- AC or DC.

Low resist / current zero

- AC ckt only
- Arc resist kept low until current is zero where arc extinguishes naturally and is prevented from restriking when the vol rising across contacts.
- neutralise the ionised particles
- replace ionised particles by un-ionised particles

RRRV

- rate of rise of re-striking vol. which decides whether the arc will re-strike or not

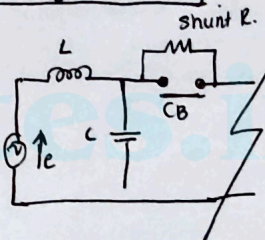
Current chopping

- phenomenon of current interruption before the natural current zero is reached. I_{no} - magnitude of current
- mainly occur in AB-CB - when breaking low currents
- Result in high voltage transient across the contacts.
- chop at i , $\therefore$ energy stored in inductance $\frac{Li^2}{2}$ is transfered to capacitance $C \rightarrow$ charging to 'e'

$$\frac{1}{2} Li^2 = \frac{Ce^2}{2}$$

- another cause is when interrupting capa current $V_g \propto I \log_b 90^\circ$

Resistance switching



- Surges can prevent by vx of shunt R connected across CB.
- Known as resist. switching

- When CB begins open - arc struck - part of arc I flow $\rightarrow R$ - result in ↓ arc current $\rightarrow$ ~~rest~~ further ↑ in arc resistance will res I to the R

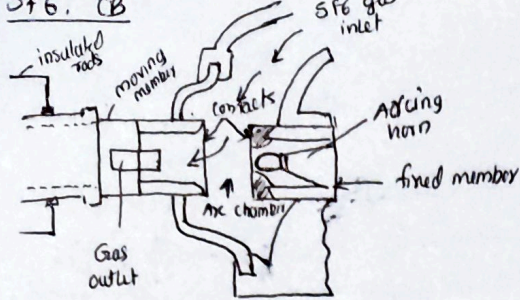
- This process continues until arc current = 0.

CB - ratings

- operate under all conditions
- under fault $\rightarrow$ 1. capable of opening fault ckt & breaking current
- 2. capable of carrying fault current $\rightarrow$ short time
- 3. capable of on after fault.

3-ratings

1. Breaking
2. Making
3. Short time

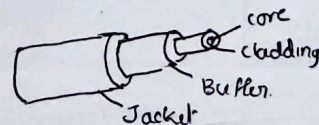


- SF₆ - highly electronegative gas - strong tendency to absorb free electrons
- Consist - FC & MC enclosed in a chamber containing SF₆ gas.
- The MC is connected to a valve control mechanism, i.e., when MC moves - valve of SF₆ reservoir opens
- Arc horn - spark quenching device

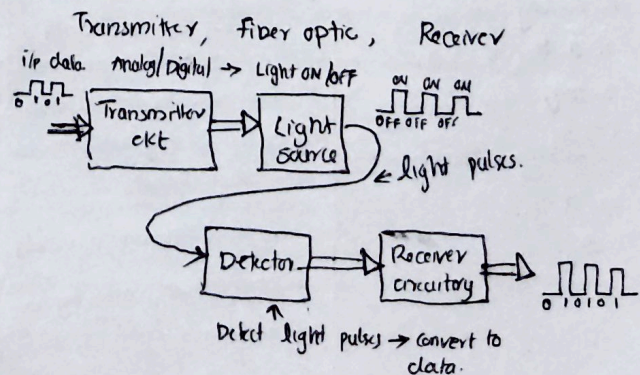
Working

Fiber optic communication

- Transmitting data using light,
- light pulse to transmit data from one point to another through an optical fiber.
- used in telephones, the internet, TV
- it uses total internal reflection principle.
- fiber optics are long, thin strands of very pure glass about size of human hair.

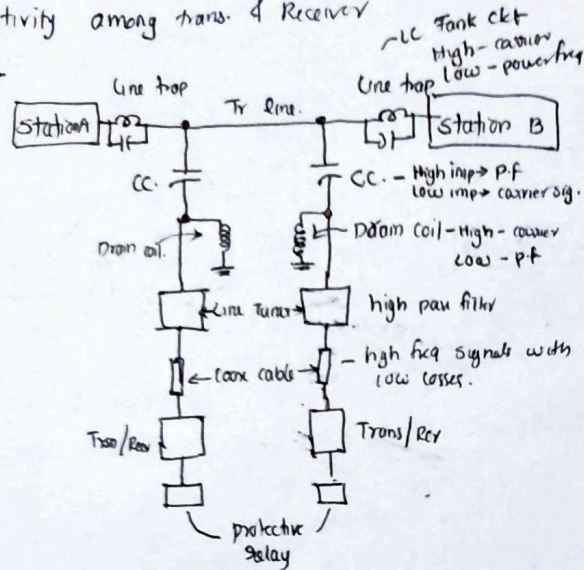


main parts:



PLCC

- The PLCC use existing electrical power lines as its commu. medium. and eliminates the need for installation of wires in order to enable connectivity among trans. & Receiver



- Trans. & distribution network.
- Home control & automation
- Entertainment
- Telecommunication
- security s/m

Limitation

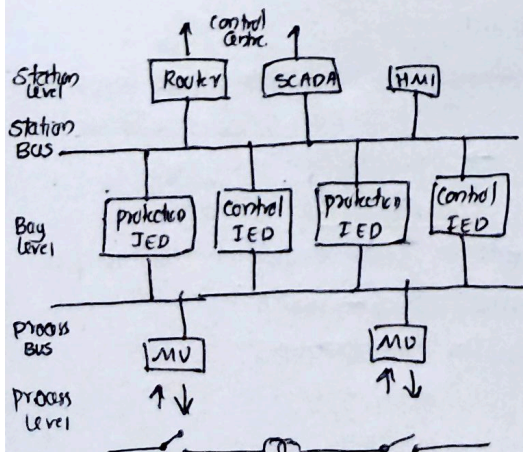
- It requires high signal
- Not secure.

Advantages

- Reliability
- cost effective

IEC 61850 standard

- It is an Ethernet based global std for communications & s/m architecture in substation automation and power distribution s/m
- It is defined by ANSI & IEC



1. Process level: - CT/PT, MU,

- Vol & current signals acquired by CT/PT are digitized by MU and set to bay level.

2. Bay level: - MP based relays, & bay controllers

IED → Intelligent Electronic Devices.

- It conduct elaborate calculations & send decision signal to other bays.

3. Station level: - HMI - (Human machine Interface) and SCADA s/m.

- status data are available to operators for monitoring and operation purposes.
- and send signals to manual controls.

Process Bus : It enables the time critical comm. b/w process level & bay level.

Station bus : exchanging information b/w bay level & station level.

Fundamental Requirements of Relays.

1. Selectivity -
2. Speed -
3. Sensitivity -
4. Reliability - ability to operate under predetermined condition
5. Simplicity
6. Economic

Electromechanical Relays.

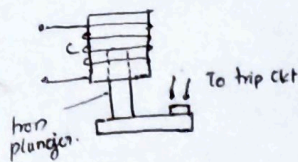
1. Electromagnetic attraction relays:

Armature being attracted to the poles of an electromag. or a plunger being drawn into a solenoid.

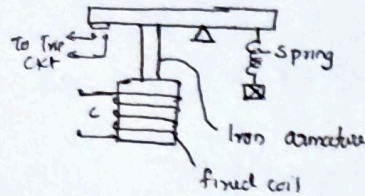
a. Attracted Armature type



D. Solid state

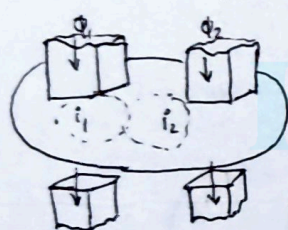


C. Balanced relay type.



2. Electromagnetic Induction Relays

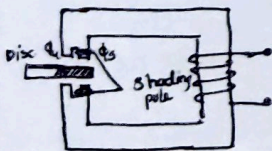
- operates on the principle of IM & are widely used for protective relaying purposes involving AC quantities.
- Not used in DC



- pivoted Al disc placed in two alternating m.f. of the same frequency but displaced in time & space

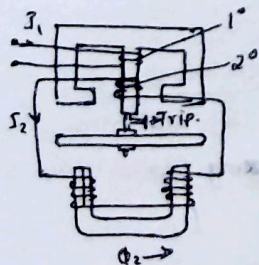
The torque produced in the disc by the interaction of one of the m.f. with currents induced in one disc by the other

a. shaded pole structure.

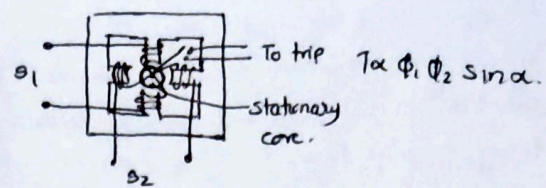


$$T \propto \Phi_s \Phi_u \sin \alpha$$

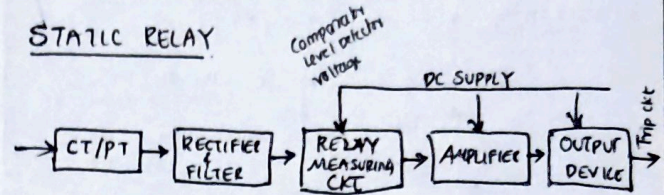
b. Watthow meter structure.



C. induction cop structure



STATIC RELAY



- Does not contain any moving parts

Advantages:

1. consumes less power
2. Quick response
3. Reset time is very less
4. high sensitivity
5. Accuracy
6. unwanted tripping is less.

Disadvantages:

1. sensitive to electrostatic field.
2. Easily affected → high vol. surges
3. less Overloading capa
4. Costly

Application

1. EHV.

Comparison and Duality

1. Comparator:

- Brain box of relay.
- sense any change in ilp
- and react quickly
- Two methods

a. Amplitude comparison:

The comparator produce an o/p signal whose amp. litude is $\propto$ the amplitude difference b/w the ilp quantities → irrespective of phase angle.

b. phase comparison:

compares the phase angles of the ilp quantity and produces pulses → width is $\propto$ to phase diff. b/w the ilp quantities

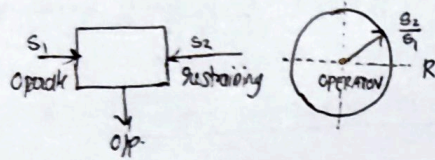
- Compares the magnitude of 2 i/p quantities irrespective of angle b/w them.

- one of i/p $\rightarrow$ operating quantity
other $\rightarrow$ restraining quantity

- Amplitude of operating qty $>$ of restraining qty, the relay sends a tripping signal to CB.

$S_1 \rightarrow$ operating $S_2 \rightarrow$ restraining

The amplitude comparator gives +ve o/p only if $\frac{S_2}{S_1} \geq k$



Phase Comparator:

- widely used $\rightarrow$ directional, distance, differential.

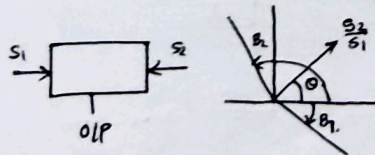
- Two i/p sig. S_1 & S_2 , the o/p occurs when one i/p's have phase difference within specified limits

- Both i/p must exist for an o/p to occur

- Independent on magnitude

- dependent $\rightarrow$ only - phase difference.

- Compares 2 qty in phase angle irrespective of magnitude & operates if the phase angle b/w them is $< 90^\circ$



operation range; $-\beta_1 \leq \theta \leq \beta_2$

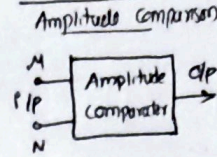
- θ is angle by which S_2 leads S_1

- If $\beta_1 = \beta_2 = 90^\circ$ the comparator is known as cosine comparator.

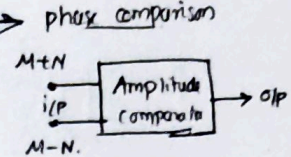
- If $\beta_1 = 0, \beta_2 = 180^\circ \rightarrow$ Sine comparator.

1. Vector product comparator

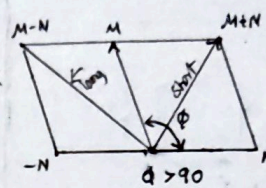
2. Coincidence type



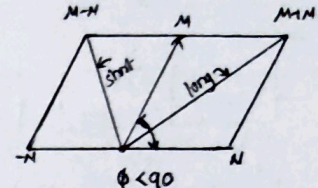
Operates when $|M| > |N|$



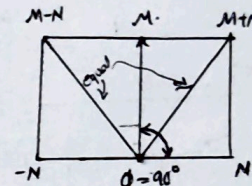
Operates when $|M+N| > |M-N|$
Satisfies when $\phi < 90^\circ$



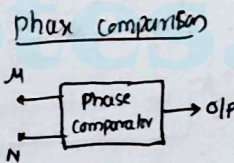
$$|M+N| < |M-N|$$



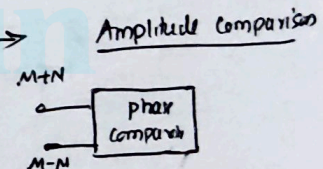
$$|M+N| > |M-N|$$



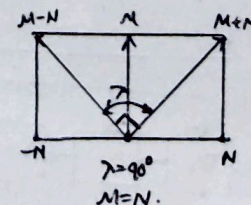
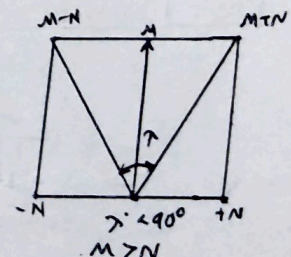
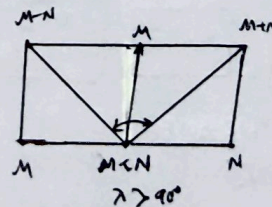
$$|M+N| = |M-N|$$



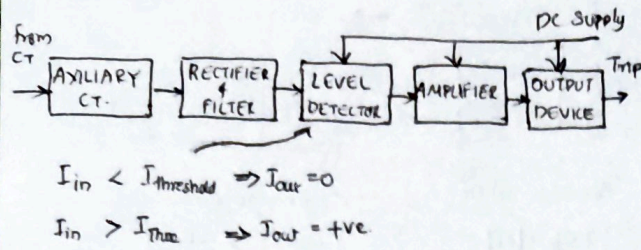
Operates when $\phi < 90^\circ$



Operates when $\lambda < 90^\circ$
Satisfies when $|M| > |N|$

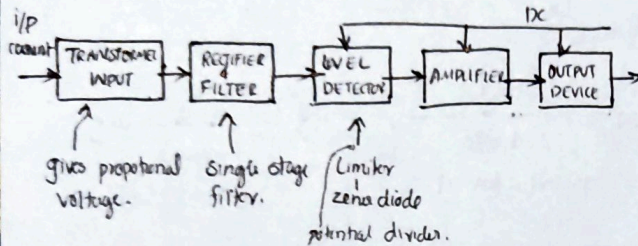


STATIC OVERCURRENT RELAY

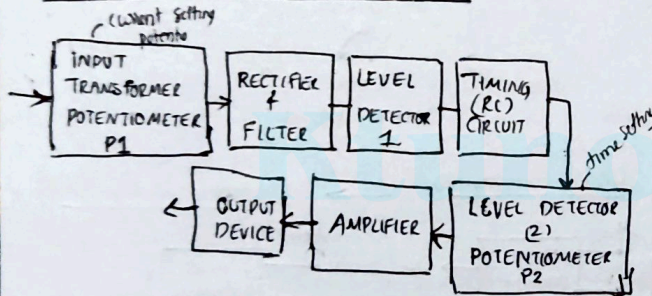


Types.

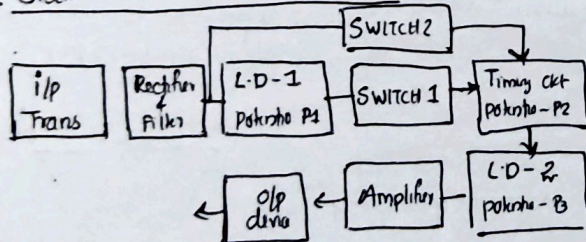
1. Static Instantaneous over current relay



2. Static Definite Time over current



3. Static Inverse time over current

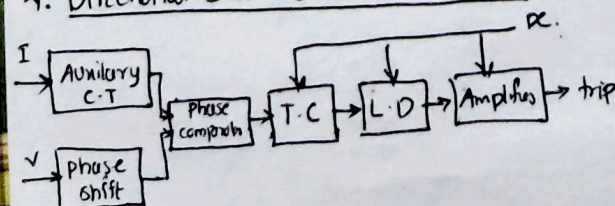


→ Normal cond. $\Rightarrow S_1$ ON ; S_2 OFF

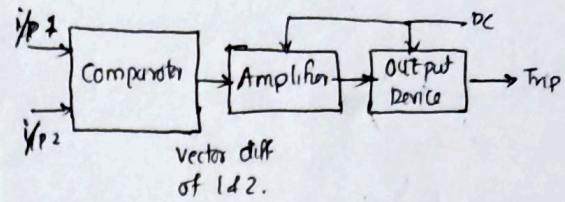
→ $I_{io} > I_T \Rightarrow S_1$ OFF ; S_2 ON.

→ Capacitor $\rightarrow$ voltage ; if exceeds setted vol. then signal is setted to ampli.

4. Directional Static over current relay



STATIC DISTANCE RELAY

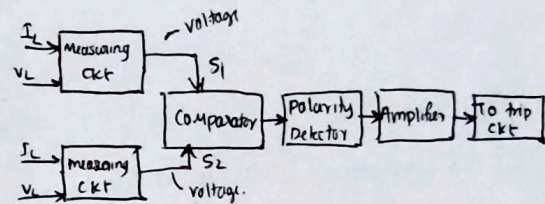


- fault occurs, comparator compare $\rightarrow$ phase difference b/w 2 i/p's $\rightarrow$ output signal will be generated.

- procc: for generators, transformers $\Rightarrow$ internal fault

- very compact, high sensitivity, stability, low power

Static Distance [or Impedance] relay



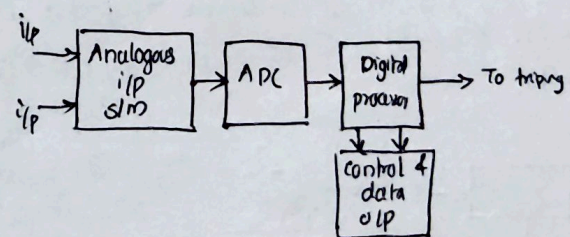
- Operation is dependent on the ratio of the voltage which is expressed in terms of impedance.

$$S_1 = k_1 I_L + k_2 V_L$$

$$S_2 = k_3 I_L + k_4 V_L$$

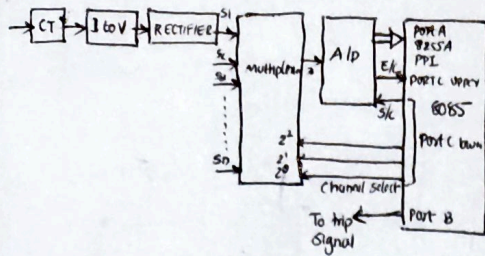
S_1 & S_2 are produced depending upon V_L & I_L chara.

Microprocessor based digital relays.



Advantages.

1. They have very small burden on CT's & PT's
2. They can be programmed.
3. very efficiently, accurately
4. one MP $\rightarrow$ several s/m
5. can predict the false or trip



- used for protection of distri lines, indu motor & equip
- using a multiplexer, the MP can sense the fault cause of a no. of ckt
- current signal from CT $\rightarrow$ proportionate vol using converter
- The AC voltage is $\rightarrow$ DC voltage using rectifier
- The voltage signal is pass to multiplexer and the out signal is fed to a A/D converter.
- The A/D convert analog voltage to a digital form then fed to MP.
- The MP sense the voltage and compare with the pre-settel or programmed value the gives a desired ofp signal.
- The MP

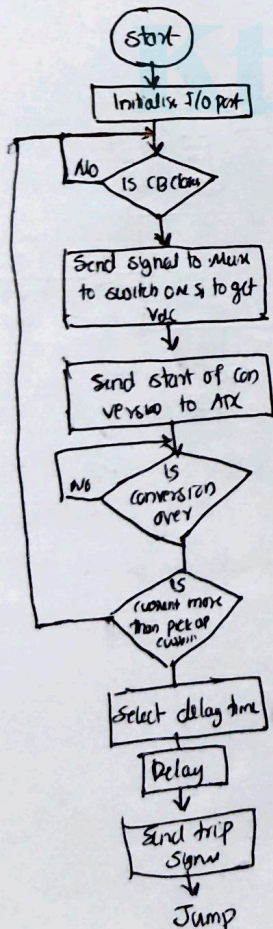
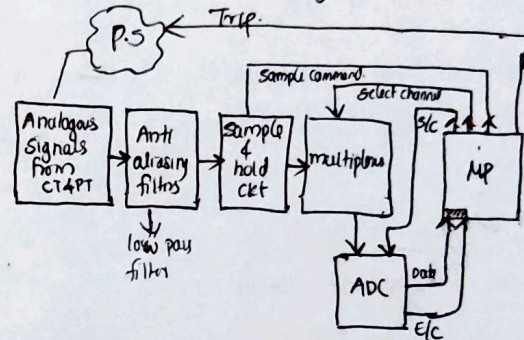
- Evolved form of a static & electromagnatic relay

- Basically a device used for measuring electric parameters in an electrical network and convert them into numerical data which undergoes mathematical & logical operations to decide on tripping an elec- network.

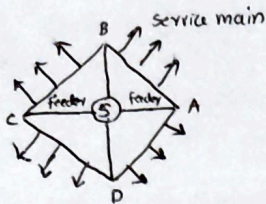
- It can monitor ~~monit~~ multiple parameters like I, V, f, onset time, offset time etc in a single ^{numerical} relay

- same relay can used for different faults.

- S/W $\rightarrow$ backbone of relay.



1. Distribution s/m.



→ Feeders - connect sub-station to area where power distributed.

→ Distributor - conductor which supply tapings are taken for supply. AB, BC, CD DA.

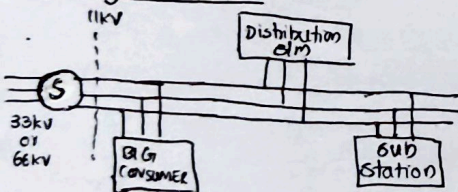
→ Service mains - small cables which connects distributor to meter.

Classification

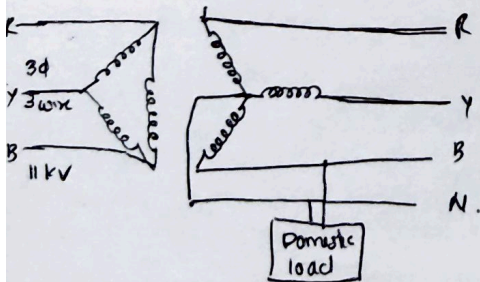
1. Nature of current - AC or DC
2. Type of construction - UG or OH.
3. Scheme of connection - Radial, Ring main, Inter-connected s/m.

AC distribution - Classified into 2 ⇒ 1st → 2nd.

i. primary distribution



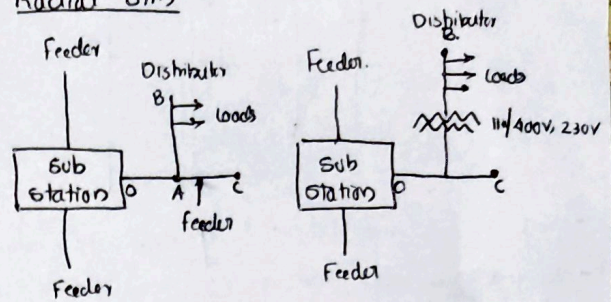
2nd distribution



- 400/230V 3φ 4 wire s/m

- 1st → 2nd →
11kV → 400V →

1. Radial s/m



DC radial s/m

AC radial s/m

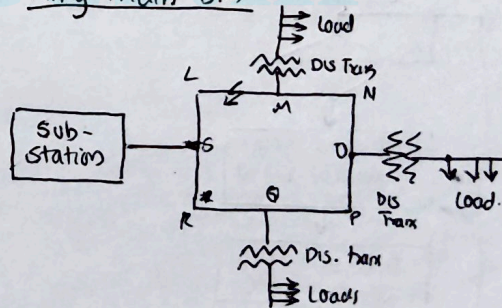
- Separate feeders radiate from a single sub-station and feed the distributors one end only.

- Simplest distri. ckt, low initial cost,

Draw backs

1. consumer depend on single feeder & distributor any fault → cut supply off.
2. end of consumer at end of distributor → voltage fluctuations

2. Ring main s/m



- 1st Distri. transformers form a loop.

- loop ckt start from sub-stations bus bar, makes a loop through area to be served and return back to the sub-station.

Advantages

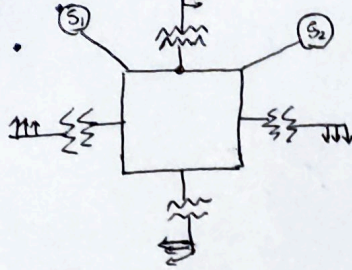
1. less vol. fluctuations
2. very reliable

3. Inter connected s/m

- The feeder ring is fed by 2 or more g/s or s/s

- Advan.

less efficiency of s/m.

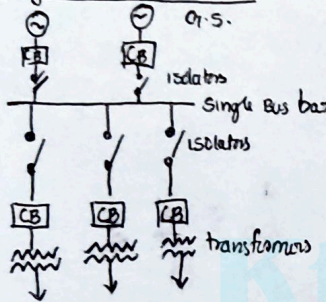


Requirements of Distri s/m

1. proper voltage.
2. Availability of power on demand
3. Reliability.

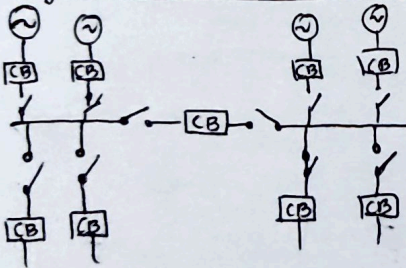
Bus bar arrangements

1. Single bus bar s/m



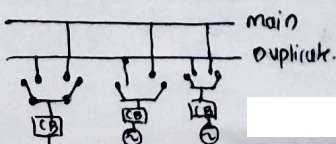
- low initial cost
 - less maintainance
 - simple operation
- mainly used in P-S

2. Single bus bar s/m with sectionalisation

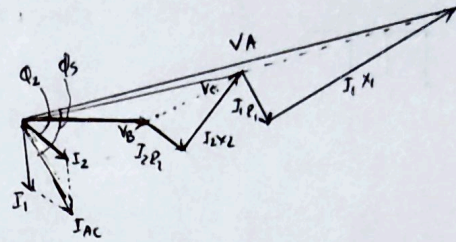
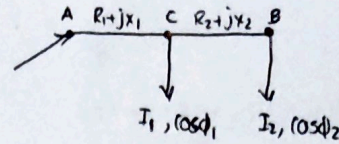


- In large G-S where several units are installed, it is common to sectionalise the bus.
- So if any fault on any section of the bus will not cause complete shut down.
- we can isolate the faulty section from bus bar arrangement.
- Easy repairing & maintainance in any section.

3. Duplicate bus-bar s/m



1. P.F. referred to V_E .



$$\bar{Z}_{AC} = R_1 + jX_1$$

$$\bar{V}_{CB} = \bar{I}_2 \cdot \bar{Z}_{CB}$$

$$\bar{Z}_{CB} = R_2 + jX_2$$

$$\bar{V}_{AC} = \bar{I}_1 \cdot \bar{Z}_{AC}$$

$$\bar{I}_1 = I_1 (\cos \phi_1 - j \sin \phi_1)$$

$$\bar{V}_A = \bar{V}_B + \bar{V}_{CB} + \bar{V}_{AC}$$

$$\bar{I}_2 = I_2 (\cos \phi_2 - j \sin \phi_2)$$

$$\bar{I}_A = \bar{I}_1 + \bar{I}_2$$

$$\bar{I}_{CB} = \bar{I}_2$$

$$\bar{I}_{AC} = \bar{I}_1$$

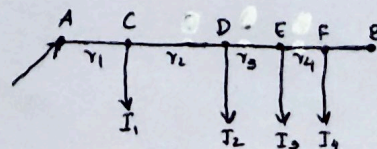
Types of DC distributors

1. Distributor fed at one end.
 2. Distributor fed at both end
 3. Distributor fed at the centre.
 4. Ring distributor.
- for concentrated & distributed loads.

Concentrated loads → load which are acting at particular points of the distributor

Distributed loads → loads which are spread over the particular distance of the distributor.

1. Fed at one end. — concentrated.



$$I_A = I_1 + I_2 + I_3 + I_4$$

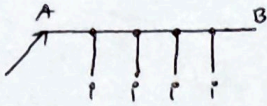
$$I_{CD} = I_2 + I_3 + I_4 \quad I_{DE} = I_3 + I_4 \quad I_{EF} = I_4$$

$$V_{AC} = \gamma_1 \cdot I_A; \quad V_{CD} = \gamma_2 \cdot I_{CD}; \quad V_{DE} = \gamma_3 \cdot I_{DE}$$

$$V_{EF} = \gamma_4 \cdot I_4.$$

$$\text{Total voltage drop} = V_{AC} + V_{CD} + V_{DE} + V_{EF}$$

Fed at one end - uniformly.



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